5	**Inferential Statistics**
5.1	• Inferential Statistics - Types Of Statistics • What Is Inferential Statistics? • What Is Hypothesis Testing? • Null Hypotheses And Alternate Hypotheses Work
5.2	• What Is P Value And Confidence Interval, Critical Value • Hypothesis Testing - Type I And Type II Errors Work
5.3	• Z-Test (One-Sample & Two-Sample) Work
5.4	• T-Test (One-Sample & Two-Sample) Work
5.5	• Central Limit Theorem • Chi-Square Test • Test Implementation Of Categorical Data Work
5.6	• ANOVA Test • Test Implementation Of Continuous Data

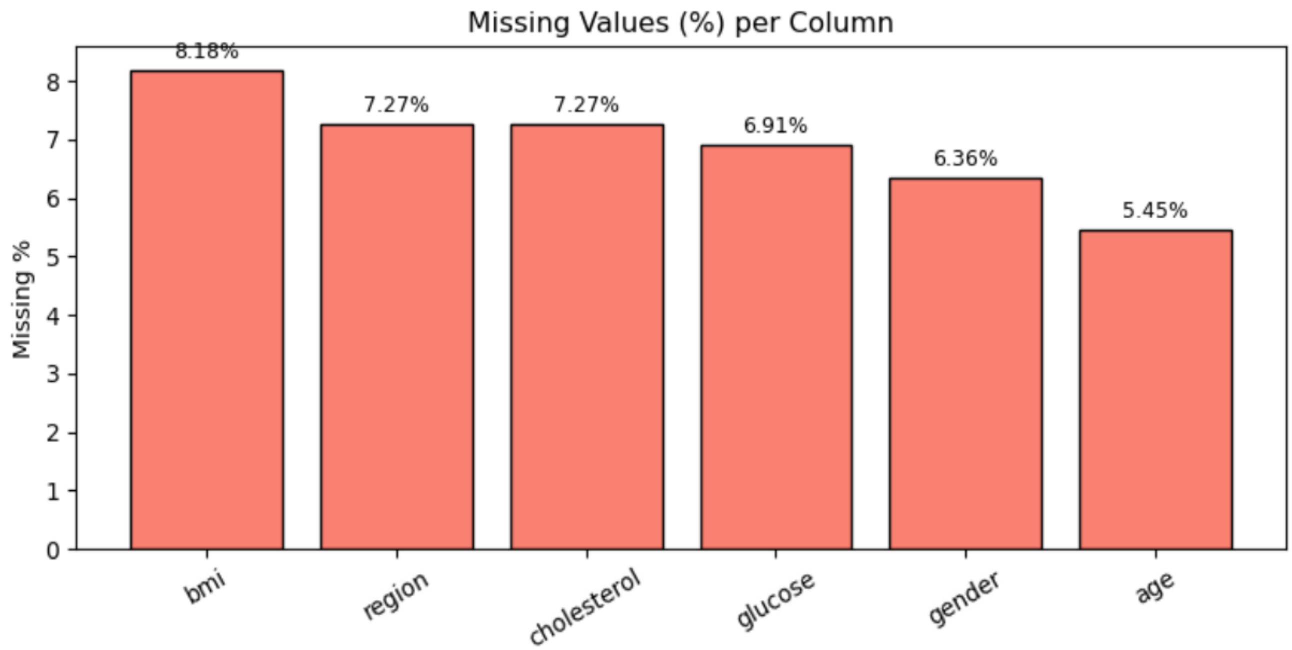
```
Dataset shape: (550, 9)
```

	patient_id	age	gender	region	bmi	blood_pressure	cholesterol	glucose	disease_risk
0	P0001	58.0	Female	East	30.04	95.9	179.2	71.3	0
1	P0002	71.0	Female	West	22.49	112.0	206.8	83.0	1
2	P0003	48.0	Female	North	21.72	110.0	204.3	90.5	1
3	P0004	34.0	Male	West	27.12	131.7	199.0	67.9	1
4	P0005	62.0	Female	South	25.34	104.8	163.2	NaN	1
5	P0006	27.0	Male	North	30.67	108.1	NaN	106.1	0
6	P0007	80.0	Female	South	28.69	113.0	257.2	119.3	0
7	P0008	40.0	Male	West	29.00	110.5	195.2	87.9	0
8	P0009	58.0	Female	West	NaN	135.8	165.7	107.9	0
9	P0010	77.0	NaN	South	NaN	103.2	NaN	62.5	1

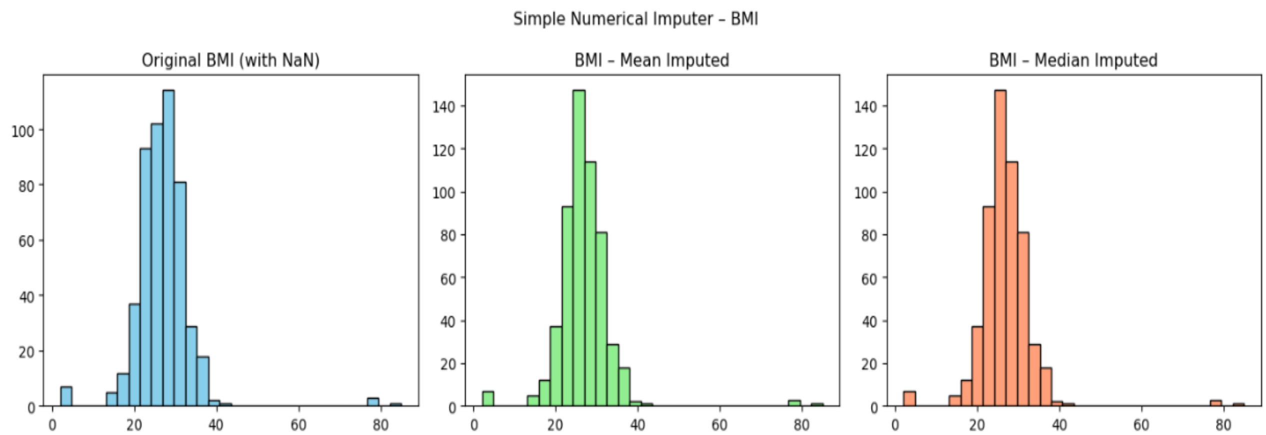
```
print(df.info())
print()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550 entries, 0 to 549
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   patient_id      550 non-null    object 
 1   age             520 non-null    float64
 2   gender          515 non-null    object 
 3   region          510 non-null    object 
 4   bmi             505 non-null    float64
 5   blood_pressure  550 non-null    float64
 6   cholesterol     510 non-null    float64
 7   glucose         512 non-null    float64
 8   disease_risk    550 non-null    int64  
dtypes: float64(5), int64(1), object(3)
memory usage: 38.8+ KB
..
```

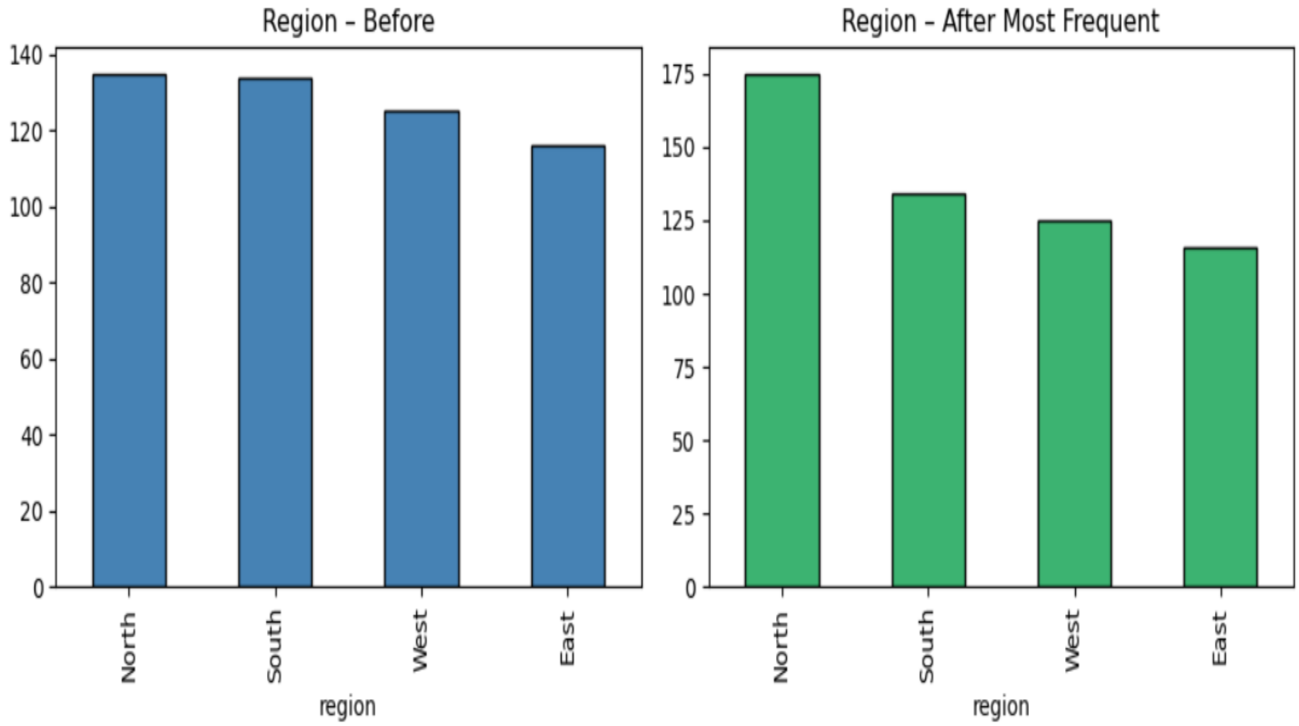
	Missing Count	Missing %
bmi	45	8.18
region	40	7.27
cholesterol	40	7.27
glucose	38	6.91
gender	35	6.36
age	30	5.45



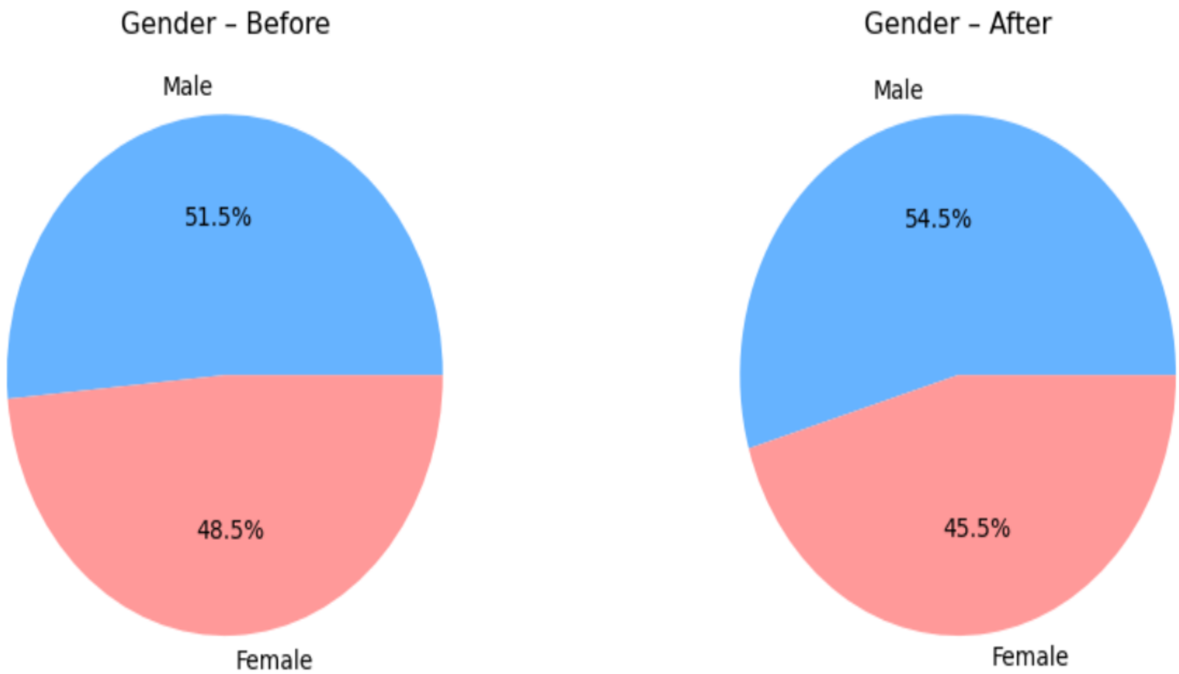
```
Rows where BMI was missing (first 10):
    patient_id  bmi  bmi_mean  bmi_median
8        P0009  NaN  26.91804       26.85
9        P0010  NaN  26.91804       26.85
23       P0024  NaN  26.91804       26.85
33       P0034  NaN  26.91804       26.85
52       P0053  NaN  26.91804       26.85
57       P0058  NaN  26.91804       26.85
66       P0067  NaN  26.91804       26.85
77       P0078  NaN  26.91804       26.85
104      P0105  NaN  26.91804       26.85
130      P0131  NaN  26.91804       26.85
```



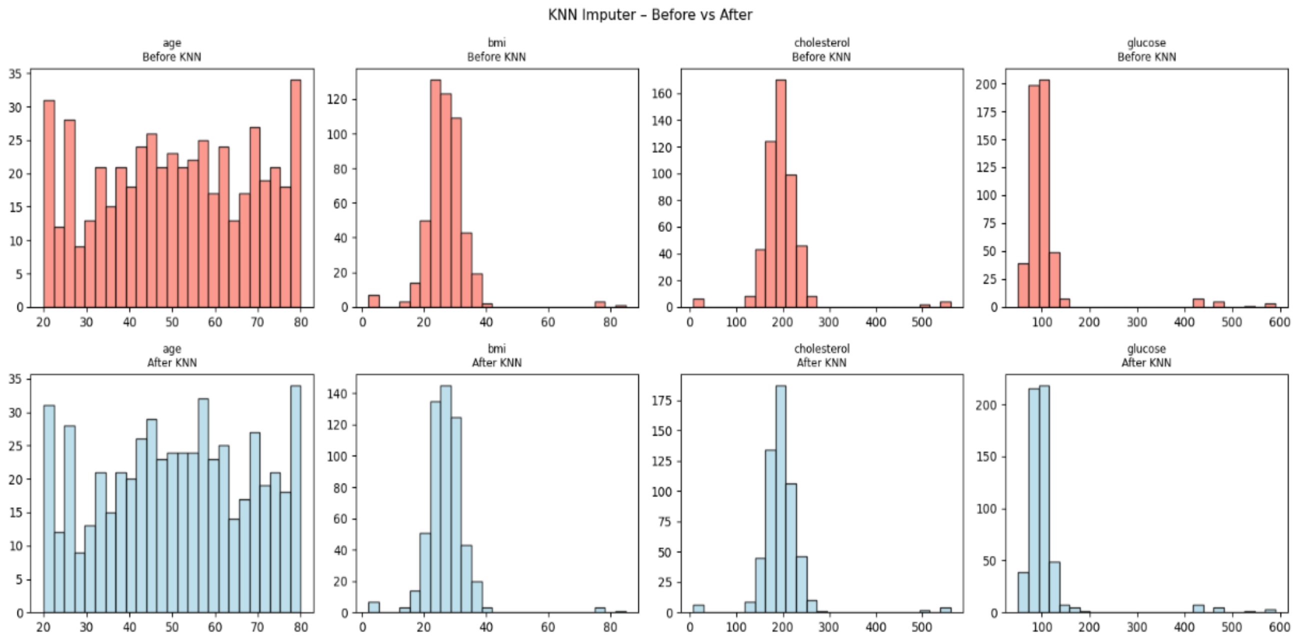
```
Most frequent region used: North
Missing before: 40
Missing after:  0
```



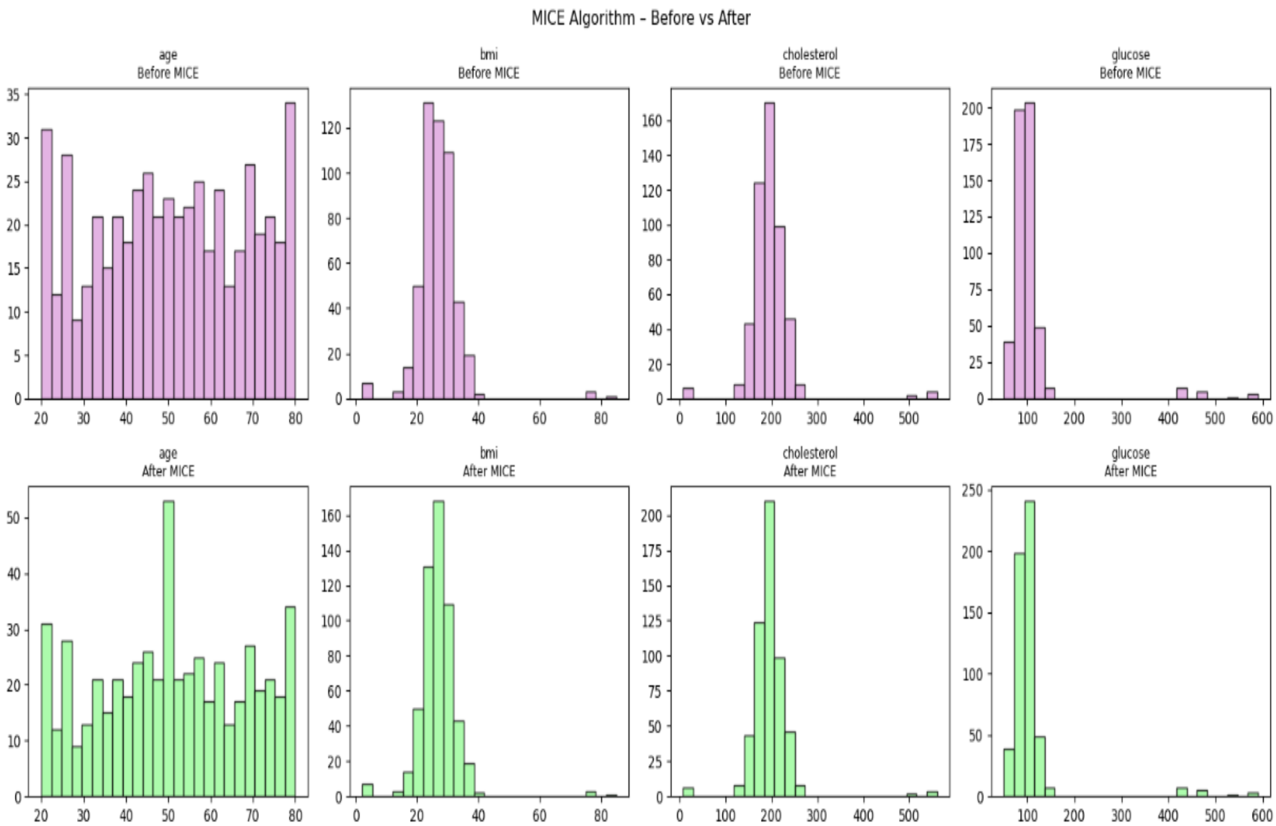
```
Most frequent gender: Male
Missing before: 35
Missing after:  0
```

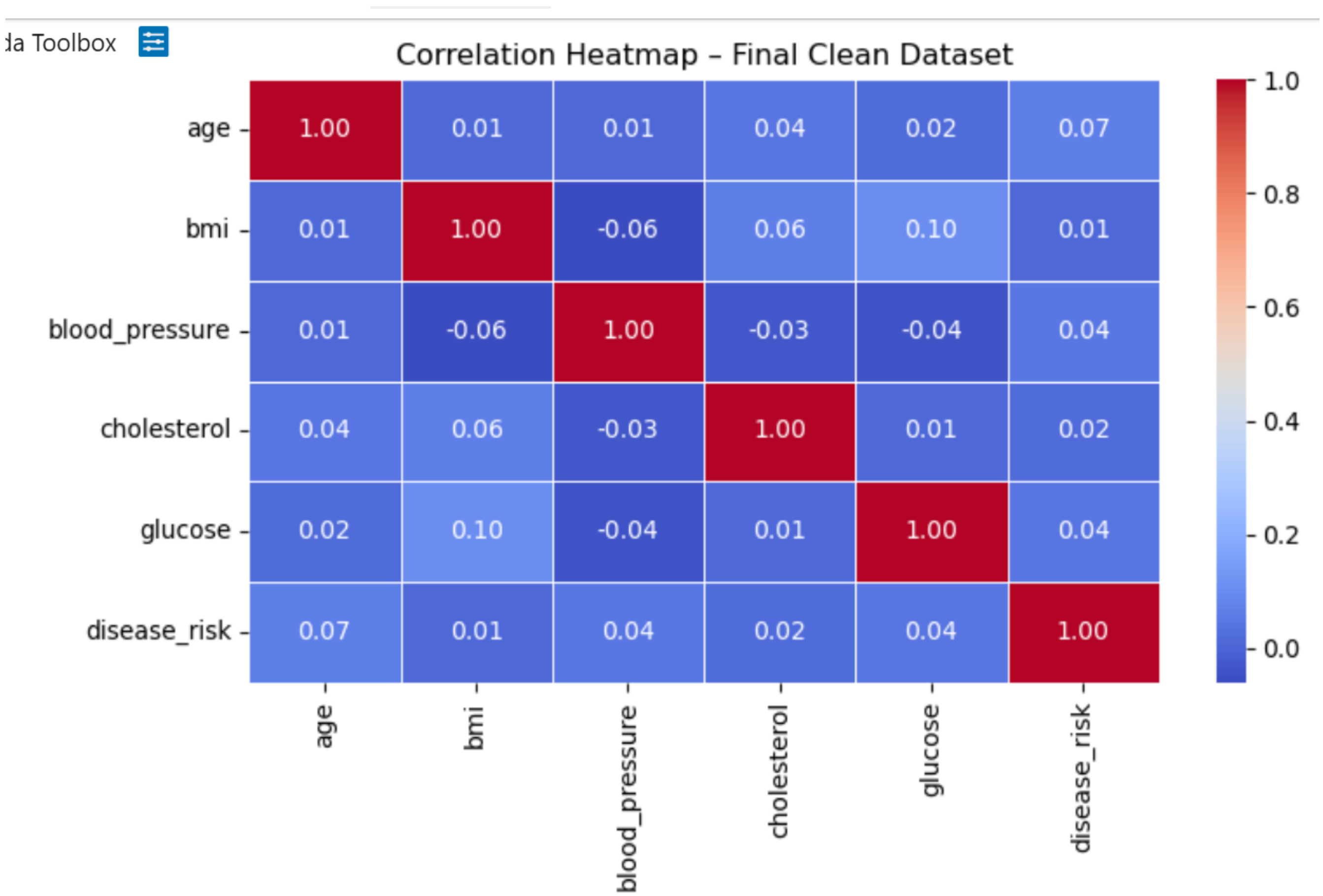


```
Missing after KNN:
age            0
bmi            0
cholesterol    0
glucose        0
dtype: int64
```



75%	65.00	29.54	211.65	107.02
max	80.00	85.00	560.00	590.00





da Toolbox
Correlation Heatmap – Final Clean Dataset
age
bmi
blood_pressure
cholesterol
glucose
disease_risk
1.00 0.01 0.01 0.04 0.02 0.07
0.01 1.00 -0.06 0.06 0.10 0.01
0.01 -0.06 1.00 -0.03 -0.04 0.04
0.04 0.06 -0.03 1.00 0.01 0.02
0.02 0.10 -0.04 0.01 1.00 0.04
0.07 0.01 0.04 0.02 0.04 1.00
1.0
0.8
0.6
0.4
0.2
0.0